Goal: Be able to understand the notation $\sum_{i=1}^{N} a_i$ and approximate the (signed) area beneath a curve.

Sigma (Summation) Notation

We want shorthand for sums with a predictable pattern. We use the Greek letter Sigma (Σ) to do so.

$$\sum_{i=m}^{n} a_i \overset{\text{def}}{=} a_m + a_{m+1} + \dots + a_n$$

"Index" → $i=m$ ← where to start; a_i: i^{th} term; n: ending value

Example 1. Compute $\sum_{i=1}^{4} i^2$

$$\sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2$$
$$= 1 + 4 + 9 + 16$$
$$= 30$$

Example 2. Compute $\sum_{j=0}^{6} (2^j + 1)$

$$\sum_{j=0}^{6} (2^j+1) = (2^0+1) + (2^1+1) + (2^2+1) + (2^3+1) + (2^4+1) + (2^5+1) + (2^6+1)$$
$$= 2 + 3 + 5 + 9 + 17 + 33 + 65$$
$$= 134$$

Example 3. Rewrite $2^3 + 3^3 + 4^3 + \dots + n^3$ in sigma notation.

$$\sum_{i=2}^{n} i^3 = 2^3 + 3^3 + 4^3 + \dots + n^3$$

Remark (Re-indexing)**.** You can start from any integer if we adjust our formula accordingly

$$\sum_{i=2}^{n} i^3 = \sum_{j=1}^{n-1} (j+1)^3 = \sum_{j=0}^{n-2} (j+2)^3 = \sum_{j=-1}^{n-3} (j+3)^3$$

Replace i with $j+1$

$i = j+1$

Example 4. Express $1^2 + 2^2 + (\sqrt{2}+1)^2 + (\sqrt{3}+1)^2 + \ldots + (\sqrt{n}+1)^2$ using sigma notation.

$$\sum_{k=0}^{n} (\sqrt{k}+1)^2$$

Theorem. Sums are linear in the sense that if k is a constant (i.e., not a function of the index), then:

(i) $\displaystyle\sum_{i=m}^{n} (a_i \pm b_i) = \sum_{i=m}^{n} a_i \pm \sum_{i=m}^{n} b_i$

(ii) $\displaystyle\sum_{i=m}^{n} k a_i = k \sum_{i=m}^{n} a_i$

Example 5. Compute $\displaystyle\sum_{i=1}^{n} k$ for any constant k.

$$\sum_{i=1}^{n} 1 = \underbrace{1+1+1+\ldots+1}_{n} = n$$

$$\sum_{i=1}^{n} k = k \sum_{i=1}^{n} 1 = kn$$

Example 6. Show $\sum_{i=1}^{n} i = 1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}$

$$S = 1 + 2 + 3 + \cdots + (n-1) + n$$
$$+\, S = n + (n-1) + (n-2) + \cdots + 2 + 1$$
$$2S = \underbrace{(n+1) + (n+1) + \cdots + (n+1) + (n+1)}_{n}$$
$$2S = n(n+1)$$
$$S = \frac{n(n+1)}{2}$$

The next theorem is not required, but it is beneficial to know since it can speed up calculations.

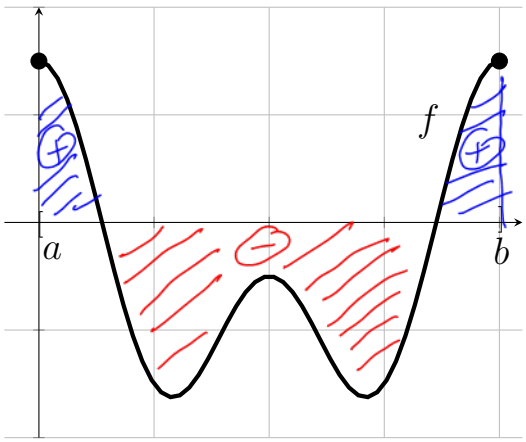
Theorem. (i) $\sum_{i=1}^{n} k = kn$ for constant k

(ii) $\sum_{i=1}^{n} i = \dfrac{n(n+1)}{2}$

(iii) $\sum_{i=1}^{n} i^2 = \dfrac{n(n+1)(2n+1)}{6}$

(iv) $\sum_{i=1}^{n} i^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

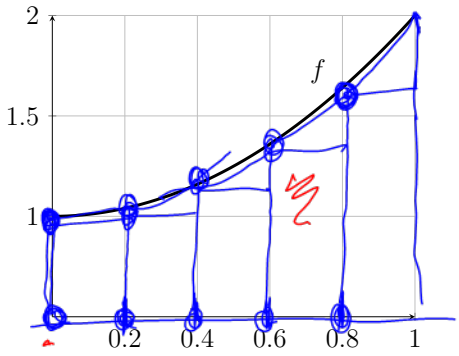
The Area Problem



Problem: Given a function f defined on a closed interval $[a, b]$, compute the (signed) area between the graph of f and the x-axis.

"Solution:" Approximate the region using rectangles.

Example 7. Approximate the area of the region beneath the graph of $f(x) = x^2 + 1$ on $[0, 1]$ using 5 rectangles.



Width: $\frac{\text{Length of } [0,1]}{5} = \frac{1-0}{5} = \frac{1}{5}$

Area $\approx \sum$ [Area of Rect.]

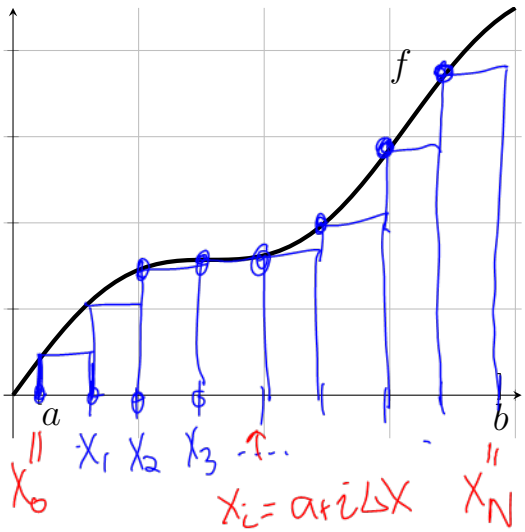
Area $\approx \frac{1}{5} f(\frac{0}{5}) + \frac{1}{5} \cdot f(\frac{1}{5}) + \frac{1}{5} \cdot f(\frac{2}{5}) + \frac{1}{5} \cdot f(\frac{3}{5}) + \frac{1}{5} \cdot f(\frac{4}{5})$

$= \sum_{i=0}^{4} \frac{1}{5} f(\frac{i}{5}) = \sum_{i=0}^{4} \frac{1}{5} \left[(\frac{i}{5})^2 + 1 \right]$

$= \frac{1}{5} \left[(\frac{0}{5})^2 + 1 + (\frac{1}{5})^2 + 1 + (\frac{2}{5})^2 + 1 + (\frac{3}{5})^2 + 1 + (\frac{4}{5})^2 + 1 \right]$

$= \frac{1}{5} \left[5 + \frac{6}{5} \right] = 1 + \frac{6}{25} = 1.24$

This is an example of a **left** Riemann Sum ("left" is referring to how we chose the height). How can we do this in general?



1. Divide $[a, b]$ into N equal regions.
2. Determine the dimensions of each rectangle.

Width: $\frac{\text{Length of } [a,b]}{N} = \frac{b-a}{N} \overset{\text{def}}{=} \Delta x$

Height: $x_i = a + i\Delta x$

Height $= f(x_i) = f(a + i\Delta x)$ for $0 \le i \le N-1$

Definition. The **left Riemann sum** of f over $[a, b]$ is defined as: ← using N rectangles

$$L_N \stackrel{\text{def}}{=} \sum_{i=0}^{N-1} f(a + i\Delta x)\,\Delta x$$

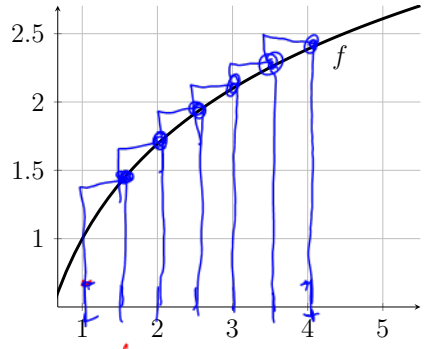
Similarly, the **right Riemann sum** is defined as:

$$R_N \stackrel{\text{def}}{=} \sum_{i=1}^{N} f(a + i\,\Delta x)\,\Delta x$$

where $\Delta x = \dfrac{b-a}{N}$

Example 8. Consider $f(x) = \ln x + 1$ on the interval $[1, 4]$: ($a = 1$, $b = 4$)

(a) Compute R_6 [i.e., the right Riemann sum with 6 rectangles].



Width: $\Delta x = \dfrac{b-a}{N}$ → $a=1$, $b=4$, $N=6$ → $\Delta x = \dfrac{4-1}{6} = \dfrac{1}{2}$

$$R_6 = \sum_{i=1}^{6} f(1 + i\cdot\tfrac{1}{2})\cdot\tfrac{1}{2} = \tfrac{1}{2}\sum_{i=1}^{6}\left[\ln\left(1+\tfrac{i}{2}\right)+1\right]$$

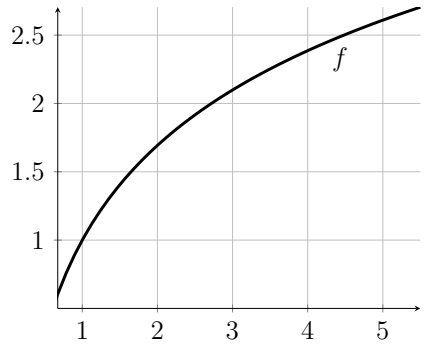
$$= \tfrac{1}{2}\sum_{i=1}^{6} 1 + \tfrac{1}{2}\sum_{i=1}^{6}\ln\left(1+\tfrac{i}{2}\right)$$

$$= 3 + \tfrac{1}{2}\left[\ln\left(\tfrac{3}{2}\right)+\ln(2)+\ln\left(\tfrac{5}{2}\right)+\ln(3)+\ln\left(\tfrac{7}{2}\right)+\ln(4)\right]$$

$$= 3 + \tfrac{1}{2}\ln\left(\tfrac{3}{2}\cdot 2\cdot\tfrac{5}{2}\cdot 3\cdot\tfrac{7}{2}\cdot 4\right)$$

$$= 3 + \tfrac{1}{2}\ln\left(\tfrac{2520}{8}\right) \approx 5.876286$$

(b) Repeat for L_6



$$\sum f(a+i\Delta x)\,\Delta x$$

(c) Set up (but do not compute) L_N and R_N.

Width: $\Delta x = \frac{b-a}{N} = \frac{4-1}{N} = \frac{3}{N}$

$$L_N = \sum_{i=0}^{N-1} \left[\ln\left(1+i\left(\tfrac{3}{N}\right)\right)+1\right]\cdot\frac{3}{N}$$

$$R_N = \sum_{i=1}^{N} \left[\ln\left(1+\tfrac{3i}{N}\right)+1\right]\frac{3}{N}$$